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Jian Chen

Senior AI Researcher

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I am a Senior AI Researcher at Dolby Laboratories working on audio foundation models, multimodal large language models, and agentic systems. My research spans language, audio, and vision, covering multimodal generation and reasoning, post-training, compression, and distillation.

EXPERIENCE

Dolby Laboratories , Sound Experience Lab — Senior AI Researcher	06/2025 – present
Adobe , Document Intelligence Lab — Research Scientist Intern	05/2024 – 11/2024
MBZUAI — Research Assistant	10/2023 – 02/2024
University at Buffalo — Research Assistant	09/2018 – 09/2025

SELECTED RESEARCH PROJECTS

Audio Foundation Models & AI Agents Full-time at Dolby Laboratories, 06/2025 — present

- Developed a transcript-free non-autoregressive voice cloning system by replacing IndexTTS2's autoregressive semantic decoder with masked generation and accelerating its flow-matching renderer, achieving 12.1× end-to-end speedup with competitive speaker similarity and intelligibility across English and Chinese benchmarks.
- Built an agentic framework for Dolby Atmos music creation, developing an LLM-based assistant that leverages tool calling to interpret user prompts and audio inputs, and orchestrates modular audio skills (e.g., source separation, spatial layout, and trajectory design) for controllable object-based spatial audio generation and editing.

LLM Compression Collaboration with Tele-AI, 02/2025 — 02/2026

- Developed **Swift-SVD**, extending KV-Core with dynamic rank allocation across layers. Showed that incremental-SVD is numerically more stable than traditional SVD-based approaches, leading to improved QA and perplexity under compression. ICML 2026 [1].
- Proposed **KV-Core**, a method for data-aware low-rank KV-cache compression in LLMs. The approach provides a simple and theoretically optimal solution by incremental-SVD, introduces NER as a compressibility metric, and benchmarks layer-wise and K/V heterogeneity in compressibility. NeurIPS 2025 Workshop [2, 3].

Multimodal Retrieval & Long-Document Understanding Internship at Adobe, 05/2024 — 11/2024

- Developed **SV-RAG**, a framework enabling multimodal LLMs for multi-page document understanding; fine-tuned Phi-3-V, PaliGemma, and InternVL2 backbones, achieving competitive performance with larger proprietary models using a 4B backbone. ICLR 2025 [4].
- Created **VisR-Bench**, a multilingual long-document benchmark for fine-grained evaluation of visual retrieval-augmented generation, exposing weaknesses in structured table reasoning and low-resource languages. [5].
- Built the **MMR** benchmark to evaluate reading and text-grounding ability in vision-language models across diverse document layouts. [6].

Multimodal Reasoning & Generative Modeling PhD student at University at Buffalo, 09/2018 — 09/2025

- Created **MusiXQA** dataset to study visual symbolic perception and cross-modal alignment for conversational music-sheet understanding; fine-tuned Phi-3-MusiX achieved strong gains over GPT baselines and showed partial robustness to real-world distortions [7].
- Developed **TextLap**, a novel model for text-based layout planning, and created the InsLap dataset for document layout generation. Fine-tuned large language models to achieve state-of-the-art performance on graphical design benchmarks. EMNLP 2024 [8]

- Contributed to the development of **LLaVA-Read**, responsible for part of the experimental work. LLaVA-Read is a multimodal large language model that uses dual visual encoders and a visual text encoder to enhance the understanding of text-rich images. [9]
- Contributed to the creation of the document dataset used for training **LaRA**, a multimodal language model that enables reading capability through OCR input. CVPR 2024 [10]
- Developed **LACE**, a continuous diffusion model using constrained optimization for layout generation and editing. It is a unified model that integrates multiple conditional inputs, achieving state-of-the-art performance across several datasets. ICLR 2024 [11]
- Developed **LRA-Diffusion**, a method for learning from noisy labels by framing data labeling as a conditional generation problem. LRA-Diffusion employs a diffusion model to learn the label generation process, using pre-trained encoders as conditional inputs to mitigate overfitting. It has consistently ranked first on three leaderboards on [Papers with Code](#). NeurIPS 2023 [12]

Machine Learning Research Research Assistant at Jacobs School of Medicine, UB, 09/2018 — 09/2022

- Designed **TimedHN**, a causal inference model capturing gene dependencies during cancer progression. The model uses binary mutation profiles as states and progression time as a latent variable, constructing a continuous-time Markov chain. An efficient optimization algorithm leveraging data sparsity was also developed to improve performance. [13]
- Proposed a neural network architecture replacing convolution with differentiable approximate string matching for microbial DNA and RNA sequence representation learning. This led to **AsMac**, an efficient large-scale sequence comparison method. [14]

EDUCATION

Ph.D. in Computer Science, <i>University at Buffalo, USA</i>	09/2018 — 06/2025
M.S. in Electrical Engineering, <i>Drexel University, USA</i>	09/2015 — 06/2017
B.S. in Applied Mathematics, <i>Hunan University, China</i>	09/2011 — 07/2015

PROFESSIONAL AND VOLUNTEER SERVICE

Peer Reviewer

- **2026** ICML, TMLR, ARR
- **2025** NeurIPS, ICLR (Notable Reviewer), ICML, AAAI, ACL, ICCV, CVPR, Genome Biology, TMLR, IJCAI
- **2024** NeurIPS, ICLR, ICML, TMLR, TETCI, PLOS ONE

RESEARCH INTERESTS

Multimodal Large Language Models, Generative Models, Representation Learning, Model Efficiency, LLM Evaluation.

SELECTED PUBLICATIONS & PREPRINTS

1. Ruoling Qi, Yirui Liu, Xuaner Wu, Xiangyu Wang, Ming Li, Chen Chen, **Jian Chen**, Yin Chen, and Qizhen Weng. Swift-SVD: Theoretical optimality meets practical efficiency in low-rank LLM compression. In *Forty-third International Conference on Machine Learning*, 2026
2. **Jian Chen**, Zhuoran Wang, Jiayu Qin, Ming Li, Meng Wang, Changyou Chen, Yin Chen, Qizhen Weng, and Yirui Liu. Kv-core: Benchmarking data-dependent low-rank compressibility of kv-caches in llms. *arXiv preprint arXiv:2602.05929*, 2026
3. **Jian Chen**, Zhuoran Wang, Jiayu Qin, Ming Li, Meng Wang, Changyou Chen, Yin Chen, Qizhen Weng, and Yirui Liu. Towards dynamic kv-cache compression: Fine-grained evaluation of key and value ranks in

llms. In *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*

4. **Jian Chen**, Ruiyi Zhang, Yufan Zhou, Tong Yu, Franck Dernoncourt, Jiuxiang Gu, Ryan A. Rossi, Changyou Chen, and Tong Sun. [SV-RAG: LoRA-Contextualizing Adaptation of MLLMs for Long Document Understanding](#). In *ICLR*, 2025
5. **Jian Chen**, Ming Li, Jihyung Kil, Chenguang Wang, Tong Yu, Ryan Rossi, Tianyi Zhou, Changyou Chen, and Ruiyi Zhang. [VisR-Bench: An Empirical Study on Visual Retrieval-Augmented Generation for Multilingual Long Document Understanding](#). *arXiv preprint arXiv:2508.07493*, 2025
6. **Jian Chen**, Ruiyi Zhang, Yufan Zhou, Ryan Rossi, Jiuxiang Gu, and Changyou Chen. [MMR: Evaluating Reading Ability of Large Multimodal Models](#). *arXiv preprint arXiv:2408.14594*, 2024
7. **Jian Chen**, Wenye Ma, Penghang Liu, Wei Wang, Tengwei Song, Ming Li, Chenguang Wang, Jiayu Qin, Ruiyi Zhang, and Changyou Chen. [MusiXQA: Advancing Visual Music Understanding in Multimodal Large Language Models](#). *arXiv preprint arXiv:2506.23009*, 2025
8. **Jian Chen**, Ruiyi Zhang, Yufan Zhou, Jennifer Healey, Jiuxiang Gu, and Changyou Chen. [TextLap: Customizing Language Models for Text-to-Layout Planning](#). In *EMNLP Findings*, 2024
9. Ruiyi Zhang, Yufan Zhou, **Jian Chen**, Jiuxiang Gu, Changyou Chen, and Tong Sun. [LLaVA-Read: Enhancing Reading Ability of Multimodal Language Models](#). *arXiv preprint arXiv:2407.19185*, 2024
10. Ruiyi Zhang, Yanzhe Zhang, **Jian Chen**, Yufan Zhou, Jiuxiang Gu, Changyou Chen, and Tong Sun. [TRINS: Towards Multimodal Language Models That Can Read](#). In *CVPR*, 2024
11. **Jian Chen**, Ruiyi Zhang, Yufan Zhou, Rajiv Jain, Zhiqiang Xu, Ryan Rossi, and Changyou Chen. [Towards Aligned Layout Generation via Diffusion Model with Aesthetic Constraints](#). In *ICLR*, 2024
12. **Jian Chen**, Ruiyi Zhang, Tong Yu, Rohan Sharma, zhiqiang xu, Tong Sun, and Changyou Chen. [Label-Retrieval-Augmented Diffusion Models for Learning from Noisy Labels](#). In *NeurIPS*, 2023
13. **Jian Chen**. Timed hazard networks: Incorporating temporal difference for oncogenetic analysis. *PLOS ONE*, 2023
14. **Jian Chen**, Le Yang, Lu Li, Steve Goodison, and Yijun Sun. [Alignment-free Comparison of Metagenomics Sequences via Approximate String Matching](#). *Bioinformatics Advances*, 2022
15. Jiayu Qin, **Jian Chen**, Rohan Sharma, Jingchen Sun, and Changyou Chen. A probability contrastive learning framework for 3d molecular representation learning. In *NeurIPS*, 2024
16. Dang Nguyen, **Jian Chen**, Yu Wang, Gang Wu, Namyong Park, Zhengmian Hu, Hanjia Lyu, Junda Wu, Ryan Aponte, Yu Xia, et al. Gui agents: A survey. In *ACL Findings 2025*, pages 22522–22538, 2025